40-question answer key and explanation guide

REVIEW DRAFT. Source-authored explanations retained; independent academic validation is pending.

01-C 02-A 03-B 04-A 05-C 06-D 07-C 08-B
09-D 10-B 11-D 12-B 13-C 14-D 15-A 16-B
17-D 18-A 19-D 20-C 21-D 22-A 23-B 24-C
25-C 26-A 27-A 28-C 29-B 30-D 31-C 32-B
33-B 34-A 35-C 36-B 37-D 38-D 39-A 40-A

1. A patient has a sudden rise in arteriolar sympathetic firing with minimal change in cardiac output. Which immediate variable rises most directly?

Answer: C. Total peripheral resistance.

Explanation: Arteriolar constriction increases resistance and therefore tends to raise arterial pressure even at stable output.

Not A: SA nodal rate concerns cardiac autonomic effects rather than arteriolar action.

Not B: Metabolite clearance depends on local blood flow, not this direct action.

Not D: Sympathetic venoconstriction generally reduces venous compliance.

Source: circulation-2:Q27 | 3, 13, 19, 21

2. The R-R separation is four large squares on a regular standard-speed ECG. What is the heart rate?

Answer: A. 75 beats/min.

Explanation: At 25 mm/s, a large square equals 0.20 s. Four large squares represent 0.8 s per beat, giving 60/0.8 = 75 beats/min.

Not B: 120 beats/min would require 2.5 large squares.

Not C: 60 beats/min would require five large squares between R waves.

Not D: 90 beats/min would require about 3.33 large squares.

Source: lecture-7:Q42 | 13-14

3. Portal hypertension promotes ascites partly by increasing which force?

Answer: B. Hydrostatic pressure in hepatic sinusoids.

Explanation: Higher sinusoidal hydrostatic pressure increases fluid filtration and hepatic lymph formation; excess can enter the peritoneal cavity.

Not A: Higher plasma oncotic pressure would oppose outward fluid filtration.

Not C: Improved lymphatic clearance would tend to oppose accumulation.

Not D: Oxygen tension is not the main Starling-force mechanism.

Source: lecture-6:Q44 | 37

4. Which contrast best distinguishes forward failure from backward failure?

Answer: A. Forward reduces delivery; backward causes venous congestion.

Explanation: Forward failure describes inadequate delivery to downstream tissues, whereas backward failure describes elevated pressure and congestion upstream of a failing chamber.

Not B: The two effects are reversed in this statement.

Not C: Reduced flow and congestion, not excessive output, are the defining patterns.

Not D: Either side can produce both forward and backward consequences.

Source: circulation-3:Q48 | Circulation 3, slides 32-33

5. What directly promotes pulmonary oedema in left-sided backward heart failure?

Answer: C. Raised pulmonary venous and capillary pressure.

Explanation: Left ventricular failure increases left atrial and pulmonary venous pressures, driving filtration into the lungs.

Not A: Lower capillary hydrostatic pressure would oppose filtration.

Not B: Systemic venous pressure is more closely linked to right-sided failure.

Not D: Right atrial pressure alone does not explain pulmonary oedema.

Source: circulation-3:Q52 | Circulation 3, slide 33; Chapter 21 ?3

6. Why does the fetal cardiovascular system divert much blood away from the pulmonary circuit before birth?

Answer: D. The placenta oxygenates blood; fetal lungs resist flow.

Explanation: The placenta performs gas exchange before birth; fluid-filled fetal lungs have high resistance, favouring right-to-left shunts.

Not A: The fetal left ventricle contributes significantly to systemic output.

Not B: Pulmonary resistance is high before aeration.

Not C: Umbilical arteries carry relatively deoxygenated blood toward placenta.

Source: lecture-6:Q53 | 42-44; Chapter 27 ?1

7. An ECG electrode detects a voltage on the skin even though the impulse originates in the myocardium. What makes this possible?

Answer: C. Cardiac electrical currents spread through conducting body tissues.

Explanation: Depolarisation and repolarisation create voltage differences that propagate through the conductive tissues of the body and can be detected at the skin.

Not A: Mechanical pulsation is distinct from electrical potential recorded by ECG electrodes.

Not B: Surface ECG electrodes do not directly measure intravascular blood pressure.

Not D: The electrodes sit on intact skin, not within the myocardium.

Source: lecture-7:Q1 | 2

8. Heart rate rises from 70 to 80 beats/min while stroke volume stays at 70 mL. By how much does cardiac output increase?

Answer: B. By 0.70 litres per minute.

Explanation: The increase is 10 beats/min x 70 mL/beat = 700 mL/min, or 0.70 L/min.

Not A: 0.80 L/min treats the extra beats as a larger volume.

Not C: 0.49 L/min does not follow from the given rate change.

Not D: 1.40 L/min doubles the actual change.

Source: circulation-2:Q20 | 1, 3

9. Most sympathetic constrictor fibres release noradrenaline onto arteriolar smooth muscle. Which receptor mainly mediates constriction?

Answer: D. The alpha-1 adrenergic receptor.

Explanation: Noradrenaline binds alpha1 receptors on arteriolar smooth muscle, raising intracellular Ca^2+ and contracting the muscle. This raises resistance and arterial pressure.

Not A: beta2 activation typically dilates muscle arterioles.

Not B: M2 receptors mediate vagal slowing of the heart.

Not C: H2 receptors respond to histamine, which dilates arterioles.

Source: circulation-2:Q25 | 13-14, 19

10. A harsh systolic murmur develops during blood ejection through a narrowed aortic valve. What mechanism explains it?

Answer: B. Turbulent forward flow through a stenotic outflow orifice.

Explanation: Aortic stenosis accelerates blood through a narrow outlet during systole, causing a turbulent ejection murmur.

Not A: The aortic valve opens when ventricular pressure exceeds aortic pressure in systole.

Not C: Mitral regurgitation usually occurs during systole, not diastole.

Not D: Laminar flow through normal valves ordinarily produces no such murmur.

Source: circulation-1:Q32 | 31-33; Chapter 5 ?7

11. At 25 mm/s, a PR interval extends across 4.5 small squares. Which interpretation is most accurate?

Answer: D. 0.18 s, inside the stated PR-interval range.

Explanation: The horizontal duration is 4.5 x 0.04 = 0.18 s. This lies within the 0.12-0.20 s interval given in the lecture.

Not A: This is not the product of 4.5 and 0.04 s.

Not B: This incorrectly halves the duration obtained from 4.5 small squares.

Not C: This incorrectly assigns 0.20 s to each small square.

Source: lecture-7:Q37 | 7, 13

12. Why does fetal haemoglobin load oxygen effectively at the placental interface?

Answer: B. Its weaker 2,3-BPG binding increases oxygen affinity.

Explanation: HbF interacts less with 2,3-BPG, shifting its oxygen dissociation curve left and favouring oxygen uptake from maternal blood.

Not A: Stronger 2,3-BPG binding would lower oxygen affinity.

Not C: Fetal haemoglobin still contains iron-bearing haem groups.

Not D: Oxygen binding is reversible, not permanent.

Source: lecture-6:Q58 | 45

13. During severe hepatic hypoperfusion, cells near the central vein are especially vulnerable to hypoxia. Why?

Answer: C. They are farthest downstream from the portal triads.

Explanation: Centrilobular cells receive blood after it has travelled from portal triads, making them less oxygenated and vulnerable.

Not A: Centrilobular hepatocytes remain within the sinusoidal perfusion system.

Not B: Oxygenated hepatic arterial inflow enters near the portal triads.

Not D: Blood oxygen content generally falls as it traverses sinusoids.

Source: lecture-6:Q41 | 35

14. Following a substantial meal, intestinal blood flow rises for several hours. What primarily drives this response?

Answer: D. Local metabolic activity and absorptive demand.

Explanation: Postprandial metabolic and absorptive activity promotes local gastrointestinal hyperaemia; the lecture notes it can last about three hours.

Not A: Sympathetic constriction usually opposes the flow increase.

Not B: Absorption increases rather than abolishes local work.

Not C: Portal flow continues to deliver absorbed substances to liver.

Source: lecture-6:Q42 | 36

15. Two Frank-Starling curves are plotted against identical end-diastolic volumes. What does an upward shift of the entire curve usually indicate?

Answer: A. Greater contractility at the same preload.

Explanation: Positive inotropy raises stroke volume or output at the same filling level, shifting the Frank-Starling function upward.

Not B: Higher afterload tends to impair ejection.

Not C: Valve closure timing is not the definition of an upward Starling curve.

Not D: Lower force would shift the curve downward, not upward.

Source: circulation-1:Q37 | 39-42

16. During a stress response, circulating adrenaline dilates vessels in working skeletal muscle while constricting many skin vessels. Why?

Answer: B. Different vascular beds express different adrenergic receptors.

Explanation: beta2 effects can favour dilation in skeletal muscle, while alpha1 effects cause constriction in many skin and gut vessels.

Not A: Muscle vessels respond because they possess adrenergic receptors.

Not C: The difference is due mainly to receptor distribution rather than selective circulating delivery.

Not D: Plasma volume differences do not explain opposite receptor responses.

Source: circulation-2:Q32 | 12-13, 30-32

17. In severe haemorrhagic shock, why does sympathetic constriction of splanchnic vessels help initially?

Answer: D. It redistributes flow and mobilises stored venous blood.

Explanation: The splanchnic bed is a large blood reservoir; vasoconstriction redirects perfusion and recruits stored blood toward central circulation.

Not A: Mucosal flow may decrease, risking ischaemia.

Not B: Extraction is not necessarily complete.

Not C: Severe or prolonged constriction can contribute to intestinal injury.

Source: lecture-6:Q43 | 36

18. An examination reveals raised JVP, a tender enlarged liver, ascites and ankle oedema. Which haemodynamic change unifies these?

Answer: A. High systemic venous pressure behind the right ventricle.

Explanation: Right ventricular backward failure elevates right atrial and systemic venous pressures, causing jugular distension, hepatic congestion and dependent oedema.

Not B: Higher systemic venous pressure promotes, rather than reduces, filtration.

Not C: Isolated pulmonary venous congestion principally reflects left-sided failure.

Not D: Right-sided backward failure increases rather than decreases venous pressure.

Source: circulation-3:Q56 | Circulation 3, slides 33

19. Immediately after the first effective breaths, what happens to pulmonary vascular resistance?

Answer: D. It falls as the lungs expand and oxygen tension rises.

Explanation: Lung aeration and increased alveolar oxygen rapidly lower pulmonary vascular resistance, increasing pulmonary blood flow.

Not A: Alveolar oxygen generally relieves pulmonary hypoxic vasoconstriction.

Not B: Pulmonary adaptation begins with the first breaths.

Not C: Pulmonary and systemic circuits retain very different resistance values.

Source: lecture-6:Q59 | 45; Chapter 27 ?4

20. A technician places V1 at the fourth intercostal space. Where should the electrode be positioned horizontally?

Answer: C. At the right sternal border.

Explanation: V1 is positioned in the fourth intercostal space at the right edge of the sternum; V2 lies at the left sternal edge at the same level.

Not A: The left midclavicular line in the fifth space is the usual site of V4.

Not B: The anterior axillary position is used for V5.

Not D: The midaxillary position is used for V6.

Source: lecture-7:Q11 | 5; Chapter 6 ?3

21. Which finding would most strongly suggest pulmonary rather than isolated dependent leg oedema?

Answer: D. Lung crackles and dyspnoea with left-heart failure.

Explanation: Left ventricular failure raises pulmonary venous pressure, causing lung interstitial and alveolar fluid with breathlessness and crackles.

Not A: Dependent leg oedema is more commonly systemic venous or local hydrostatic.

Not B: Pitting alone does not localise oedema to the lungs.

Not C: Postmastectomy swelling suggests lymphatic obstruction.

Source: circulation-1:Q91 | 90-95; Chapter 14 ?9

22. Why can longstanding filarial lymphoedema become firm and poorly pitting?

Answer: A. Chronic protein-rich interstitial fluid promotes tissue fibrosis.

Explanation: Persistent failure of lymphatic drainage leaves protein-rich fluid in the interstitium, promoting chronic inflammation and fibrosis. This explains why longstanding lymphoedema becomes firm and often non-pitting.

Not B: Rapid reabsorption would tend to reduce swelling and protein accumulation.

Not C: Chronic interstitial protein deposition can drive fibrosis despite any venous tone changes.

Not D: The chronic fibrosis is linked to sustained lymphatic obstruction, not a brief arterial-pressure change.

Source: circulation-1:Q97 | 95; Chapter 15 ?7

23. A patient with left-sided forward heart failure reports fatigue and produces very little urine. Which mechanism links the two?

Answer: B. Reduced systemic perfusion limits muscle and renal blood flow.

Explanation: Reduced blood ejected from the left heart impairs delivery to skeletal muscle and kidneys, causing fatigue and oliguria.

Not A: Higher venous pressure does not directly improve GFR.

Not C: The primary forward mechanism is reduced output, not pulmonary congestion.

Not D: The syndrome involves reduced rather than excessive systemic delivery.

Source: circulation-3:Q53 | Circulation 3, slides 32

24. A recording shows that cardiac vagal stimulation lowers the firing rate of the sinoatrial node. Which additional effect is expected?

Answer: C. Slower impulse conduction through the AV node.

Explanation: Parasympathetic acetylcholine acts at muscarinic M2 receptors to slow both SA nodal firing and AV nodal conduction.

Not A: beta1 activation describes sympathetic, not vagal, stimulation.

Not B: AV conduction is slowed rather than accelerated by vagal action.

Not D: The major cardiac pacemaker effect is not mediated by alpha1.

Source: circulation-2:Q29 | 20

25. What conservation principle underlies Fick measurement of cardiac output?

Answer: C. Oxygen use equals flow x arterial-venous oxygen difference.

Explanation: By mass balance, oxygen uptake across lungs and consumption by tissues equal cardiac output multiplied by arterial-venous oxygen content difference.

Not A: Tissues extract oxygen, so venous content is generally lower.

Not B: Extraction and flow together determine total oxygen consumption.

Not D: The body consumes oxygen but does not produce it.

Source: circulation-1:Q45 | 52-53

26. What keeps tissue fluid moving through collecting lymph vessels toward the great veins?

Answer: A. Valves, muscle movement and lymphatic contractions.

Explanation: One-way valves, skeletal-muscle compression, respiratory movement and lymphatic smooth-muscle contraction promote forward flow. The valves prevent reflux between compressions, making the low-pressure pumping effective.

Not B: Whole vessels do not diffuse through tissues.

Not C: Backflow is limited by valves rather than required for transport.

Not D: Lymph flow is not driven directly by systemic arterial pressure.

Source: circulation-1:Q94 | 60; Chapter 15 ?3

27. A patient with left-sided heart failure feels breathless when lying flat but improves after sitting up. Why?

Answer: A. Lying flat shifts peripheral blood toward congested lungs.

Explanation: Supine positioning returns peripheral blood to the thorax; a failing left ventricle cannot handle the extra flow, increasing pulmonary venous congestion.

Not B: Central thoracic volume commonly increases rather than decreases supine.

Not C: A higher pulmonary capillary pressure would worsen breathlessness.

Not D: Sitting upright tends to reduce thoracic venous return.

Source: circulation-3:Q55 | Circulation 3, slides 33; Chapter 21 ?3

28. A regular rhythm has 24 small squares between adjacent R waves at standard paper speed. Approximately what is the heart rate?

Answer: C. 63 beats/min.

Explanation: The rate in a regular rhythm is 1500 divided by the R-R separation in small squares: 1500/24 = 62.5 beats/min, approximately 63.

Not A: 50 beats/min corresponds to about 30 small squares between R waves.

Not B: 80 beats/min corresponds to about 18.75 small squares between R waves.

Not D: 60 beats/min corresponds to about 25 small squares between R waves.

Source: lecture-7:Q41 | 13-14

29. What is detected by the fingers when palpating the radial pulse?

Answer: B. An arterial wall-distension pressure wave.

Explanation: Each ventricular ejection creates an arterial pressure/distension wave that travels along the arteries. This pressure wave moves faster than the bulk blood itself.

Not A: Venous blood does not directly pulse into systemic arteries.

Not C: Red cells are not separately felt during palpation.

Not D: The palpable pulse is mechanical rather than an electrical conduction wave.

Source: circulation-1:Q66 | 22-23

30. V2 has been placed correctly and V4 is at its usual site. Where should V3 be attached?

Answer: D. Halfway between the V2 and V4 electrodes.

Explanation: V3 is placed between the already positioned V2 and V4, so the anatomical line between those contacts determines its location.

Not A: The midaxillary line at V4 level is used for V6.

Not B: The anterior axillary line at V4 level is used for V5.

Not C: The fourth right parasternal position belongs to V1.

Source: lecture-7:Q12 | 5; Chapter 6 ?3

31. Where is the largest controllable fall in systemic vascular pressure normally produced?

Answer: C. Across resistance arterioles.

Explanation: Small arteries and arterioles provide substantial resistance, producing a major pressure drop from arterial to capillary levels.

Not A: Large veins carry blood at low pressure and resistance.

Not B: Right atrial pressure is low but does not create the main distributed arterial drop.

Not D: Large elastic arteries conduct flow with relatively small resistance.

Source: circulation-1:Q65 | 57, 61, 76; Chapter 10 ?9

32. Why does the sinoatrial node ordinarily set the heart's rhythm rather than latent pacemakers?

Answer: B. It has the fastest intrinsic depolarisation rate.

Explanation: The SA node normally reaches threshold fastest and overdrive-suppresses slower latent pacemakers. Faster SA-node impulses normally suppress slower escape pacemakers.

Not A: Action potential duration is not the primary reason for the fastest spontaneous rate.

Not C: Coronary perfusion is not determined by pulmonary venous timing.

Not D: Pacemaker dominance does not depend on muscle mass.

Source: circulation-1:Q12 | 7-9; Chapter 3 ?3

33. Why can coordinated cardiac pumping continue without a new external nerve impulse for every beat?

Answer: B. Intrinsic pacemaker cells generate rhythmic impulses.

Explanation: Specialised cardiac pacemaker cells depolarise spontaneously; autonomic nerves modulate rather than initiate each heartbeat.

Not A: Atrial conduction is not limited to expiration.

Not C: The SA node produces electrical impulses, not direct ejection pressure.

Not D: Some conducting cells possess spontaneous automaticity.

Source: circulation-1:Q4 | 7-10

34. A patient has near-normal resting cardiac output but very high ventricular filling pressures and pulmonary congestion. Which statement fits best?

Answer: A. Heart failure can occur with normal output.

Explanation: Heart failure includes states where output is maintained only at abnormally high filling pressures, causing congestion.

Not B: High filling pressures can be evidence of dysfunction rather than normality.

Not C: Diastolic failure can occur with preserved ejection fraction.

Not D: A near-normal output can be maintained at harmful filling pressures.

Source: circulation-3:Q47 | Circulation 3, slide 31; Chapter 21 ?1

35. Which pressure difference primarily drives blood back to the right atrium from the body?

Answer: C. Systemic filling pressure minus right atrial pressure.

Explanation: Venous return depends on the upstream systemic filling pressure relative to right atrial pressure and the resistance to venous return.

Not A: Aortic-right ventricular systolic pressure is not the venous return driving gradient.

Not B: That difference is a pulse-pressure-related comparison.

Not D: Pulmonary venous pressure does not define systemic venous return.

Source: circulation-1:Q74 | 97; Chapter 12 ?3

36. What can follow when the right ventricle fails to deliver adequate forward flow to the lungs?

Answer: B. Less blood reaches the left heart for systemic output.

Explanation: Reduced right ventricular output limits pulmonary flow and subsequent return to the left atrium, compromising left-sided filling and delivery.

Not A: The right heart cannot deliver excess flow when severely failing.

Not C: Reduced pulmonary transit cannot raise systemic oxygen delivery.

Not D: Less pulmonary venous return tends to decrease left-sided filling.

Source: circulation-3:Q54 | Circulation 3, slides 32

37. At an otherwise unchanged capillary, how would increased interstitial protein concentration affect fluid movement?

Answer: D. It strengthens the outward oncotic force favouring filtration.

Explanation: More interstitial protein increases interstitial colloid osmotic pressure, tending to pull fluid outward from the capillary and increase filtration.

Not A: Increased proteins outside the capillary attract rather than remove interstitial water.

Not B: Plasma protein concentration is held constant; the changed force is interstitial.

Not C: A change in interstitial protein does not directly lower capillary pressure.

Source: circulation-1:Q83 | 85-88

38. Electrical stimulation of the medullary pressor area increases arterial pressure. What is the principal neural pathway?

Answer: D. Excitation of sympathetic neurons in the spinal cord.

Explanation: The pressor area sends excitatory fibres down to the sympathetic neurons in the spinal cord. More sympathetic output constricts arterioles and veins and speeds and strengthens the heart, so pressure rises.

Not A: Carotid sinus fibres carry sensory information towards the medulla.

Not B: Inhibiting sympathetic neurons would tend to lower pressure.

Not C: Pressor-area activity does not require direct pacemaker inhibition.

Source: circulation-2:Q22 | 16, 19, 21

39. A patient has low circulating volume. Which hormone chiefly increases renal water retention and can constrict vessels at high concentration?

Answer: A. Vasopressin, also called ADH.

Explanation: ADH promotes renal water conservation and has a vasoconstrictor action when its level is high.

Not B: ANP promotes sodium and water loss.

Not C: VIP is a vasodilator rather than a water-retaining hormone.

Not D: Prostacyclin dilates vessels and limits platelet aggregation.

Source: circulation-2:Q34 | 12, 29

40. Why might severe anaemia produce a cardiac flow murmur despite structurally normal valves?

Answer: A. Lower viscosity and increased flow favour turbulence.

Explanation: Anaemia lowers viscosity and may increase cardiac output, raising the tendency to turbulent flow and a functional murmur.

Not B: The murmur is not due to forced aortic valve closure.

Not C: Anaemia decreases viscosity rather than increasing it.

Not D: Conduction block is not the principal mechanism of flow murmurs.

Source: circulation-1:Q31 | 31-34